

# SV-Flow: Lightweight Inference-Time Diversity Enhancement for Structure-Based Drug Design via Kinematic Decoupling

Author Name(s)

*Institution(s)*

*Correspondence: corresponding.author@institution.edu*

## Abstract

Structure-based drug design (SBDD) generative models based on flow matching and diffusion have achieved remarkable success in generating drug-like molecules conditioned on protein binding pockets. However, these models suffer from mode collapse during independent sampling, producing highly homologous molecules clustered in pocket centers and failing to explore diverse binding modes. Existing post-hoc diversity enhancement methods are either computationally expensive (e.g., MaxMin selection from large sampling pools) or physically implausible (e.g., translational shifts followed by energy minimization).

We present SV-Flow (Stein Variational Flow Matching), a lightweight inference-time diversity enhancement framework that couples single-particle flow matching trajectories into a multi-particle variational inference system. The key innovation is **kinematic decoupling**: we decompose the velocity field into internal conformational velocity (preserving chemical integrity) and center-of-mass (CoM) translational velocity (subject to Stein Variational Gradient Descent repulsion). SVGD repulsion is applied exclusively in the  $\mathbb{R}^3$  centroid space, mathematically guaranteeing zero strain on chemical bonds. A time-annealing schedule ( $t_{\text{on}} = 0.5$ ) delays the onset of repulsion to avoid artifacts from early noise states.

On the CrossDocked2020 test set ( $N = 100$  pockets, 10 trajectories per pocket), SV-Flow Core achieves a mean Tanimoto chemical diversity of **0.882** (vs. 0.848 for DrugFlow independent sampling), representing a **4.1% relative improvement** (paired  $t$ -test:  $t(99) = 4.35$ ,  $p = 3.36 \times 10^{-5}$ , Cohen’s  $d = 0.435$ ). Centroid spatial variance increases by **25.8%** (0.126 vs. 0.100  $\text{\AA}^2$ ). SV-Flow Core with  $N = 10$  approaches the diversity of MaxMin post-hoc selection from a  $5\times$  larger sampling pool ( $N = 50 \rightarrow 10$ ; Tanimoto diversity 0.902), demonstrating that coupled SVGD repulsion during sampling is fundamentally more efficient than post-hoc filtering. The QED drug-likeness exhibits a modest trade-off (0.504 vs. 0.547 for DrugFlow,  $p = 2.41 \times 10^{-4}$ ), consistent with the expected quality-diversity Pareto frontier. Post-minimization physical validity is comparable between methods.

**Key insight:** Through systematic ablation studies, we discovered that initially designed geometric constraints (tangent-plane projection and orthogonal kinetic protection) actually **suppress spatial diversity**. Removing these constraints—pure SVGD repulsion with time annealing—restores full exploration capability while maintaining chemical integrity. This “Less is More” principle provides a concise design principle for inference-time guidance.

SV-Flow adds minimal overhead ( $\approx 0.2$  s/molecule for SVGD repulsion) and requires no retraining. The framework is model-agnostic and can be readily applied to any flow-matching or diffusion-based SBDD pipeline.

## 1 Introduction

Structure-based drug design (SBDD) aims to discover novel small molecules that bind to target proteins with high affinity and specificity. Recent advances in deep generative models have

transformed this field: flow matching and diffusion models [1, 2, 3, 4] can now generate drug-like molecules directly conditioned on protein binding pocket geometry, producing candidates with favorable predicted binding poses and chemical validity. These models learn to iteratively denoise random atomic coordinates into physically plausible ligand conformations, capturing rich geometric and chemical regularities from protein–ligand complex databases.

Despite these advances, a fundamental limitation persists: **mode collapse**. When generating multiple samples independently from the same pocket, existing flow matching models produce highly homologous molecules clustered around the pocket center [5]. This collapses the exploration of diverse binding modes—different ligands occupying distinct sub-pockets or forming alternative interaction patterns—which is crucial for discovering novel chemotypes, expanding patent space, and identifying leads with unique binding profiles and reduced off-target risks.

Several approaches have been proposed to address this challenge. Post-hoc selection methods, such as MaxMin diversity selection [23], generate a large pool of candidates and filter for diversity, but this is computationally expensive and wasteful: the majority of generated samples are discarded. Translational shifts followed by energy minimization can improve spatial distribution, but they lack the synergistic adjustment of molecular conformations during generation, leading to strained geometries that may not represent physically meaningful binding modes.

A recent and promising direction is inference-time guidance, which imposes design objectives on pre-trained generators without retraining. Building on the foundational frameworks of classifier guidance [6] and classifier-free guidance [7], the universal guidance framework [8] enables gradient-based biasing through arbitrary differentiable energy functions. In the biomolecular domain, Lam et al. [9] introduced Metadiffusion, which applies Stein Variational Gradient Descent (SVGD) repulsion during the diffusion denoising process to enhance conformational diversity. However, Metadiffusion applies repulsive forces in the full atomic coordinate space ( $\mathbb{R}^{3N}$ ), which introduces destructive kinetic energy into internal conformational degrees of freedom. For small molecules, chemical integrity is far more fragile than protein backbone conformations—isotropic repulsion at the atomic level can distort bond lengths, break chemical bonds, and create unrealistic steric clashes.

We present **SV-Flow (Stein Variational Flow Matching)**, a lightweight inference-time diversity enhancement framework that addresses these limitations through **kinematic decoupling**. The core insight is to separate the velocity field into two orthogonal components:

1. **Internal conformational velocity** ( $\mathbf{v}_{\text{int}}$ ): governs torsional rotations and bond-angle changes, fully preserving chemical integrity;
2. **Center-of-mass translational velocity** ( $\mathbf{v}_{\text{CoM}}$ ): governs global positioning within the pocket, subject to SVGD repulsion.

SVGD repulsion is applied **exclusively** in the  $\mathbb{R}^3$  centroid space, providing a mathematical guarantee of zero strain on chemical bonds. A time-annealing schedule ( $t_{\text{on}} = 0.5$ ) delays the onset of repulsion until later stages of generation, allowing chemical topology to form before imposing spatial diversity constraints.

**Key Discovery:** Through systematic ablation studies, we discovered a counter-intuitive principle—"Less is More." Initially, we designed two geometric constraints as safety barriers: tangent-plane projection (restricting repulsion to the protein surface) and orthogonal kinetic protection (filtering out velocity components parallel to  $\mathbf{v}_{\text{CoM}}$ ). However, comprehensive ablation experiments revealed that these constraints **severely suppress spatial diversity**. Removing them—pure SVGD repulsion with time annealing—restored full exploration capability ( $2.44\times$  improvement over the geometrically-constrained variant) while maintaining chemical integrity comparable to the baseline. This challenges the intuition that more constraints yield safer generation.

**Our contributions** are:

1. **Methodological:** We are the first to couple the SVGD interaction framework with flow matching ODEs for small molecule SBDD. Through kinematic decoupling, repulsion is mathematically guaranteed to have zero impact on chemical bonds.
2. **Empirical:** Systematic ablation across 5 variants reveals the “Less is More” principle: geometric constraints designed to protect the system actually suppress diversity, and pure SVGD repulsion with time annealing achieves the optimal balance.
3. **Practical:** SV-Flow achieves spatial diversity enhancement with  $O(N^2 \cdot 3)$  computational overhead, negligible compared to the  $O(N_{\text{atoms}} \cdot d^2)$  cost of the base model. The framework is model-agnostic and requires no retraining.

## 2 Related Work

### 2.1 Structure-Based 3D Molecular Generation

3D generative models for SBDD have evolved rapidly from early conditional variational autoencoders to modern diffusion and flow matching architectures. **TargetDiff** [1] pioneered equivariant denoising diffusion for pocket-conditioned ligand generation, using an SE(3)-equivariant graph neural network to jointly generate atom types and coordinates. **DiffSBDD** [2] operates on protein surface point clouds and uses a diffusion model over atomic positions. **DecompDiff** [5] decomposes molecules into arms and scaffolds for more controllable generation. **DrugFlow** [3] adopts a flow matching formulation [12] with an SE(3)-equivariant Geometric Vector Perceptron architecture, achieving state-of-the-art validity rates on the CrossDocked benchmark. **FLAG** [4] extends flow matching to full-atom ligand generation with explicit hydrogens.

A well-documented but rarely directly addressed limitation across these models is mode collapse: when sampling multiple trajectories from the same pocket, molecules tend to cluster in chemically similar regions, failing to explore diverse binding modes. This is a consequence of the independent sampling paradigm—each trajectory is generated in isolation, with no mechanism to encourage inter-trajectory diversity.

### 2.2 Inference-Time Guidance for Generative Models

Inference-time guidance has emerged as a flexible paradigm for imposing design objectives on pre-trained generative models without retraining. Classifier guidance [6] uses gradients from an auxiliary classifier to steer diffusion model outputs toward desired class labels. Classifier-free guidance [7] achieves conditional control without a separate classifier by interpolating between conditional and unconditional score estimates. Universal guidance [8] generalizes these approaches, enabling gradient-based biasing through arbitrary differentiable energy functions.

In the molecular domain, this paradigm has been adopted in several forms. **Metadiffusion** [9] applies SVGD repulsion [10] during the denoising process of the Boltz-2 biomolecular diffusion model, using a meta-energy function to steer conformational ensembles toward experimental observables (SAXS, NMR) and user-specified collective variables. While effective for protein conformation generation, Metadiffusion operates in the full atomic coordinate space ( $\mathbb{R}^{3N}$ ), which is incompatible with the rigid constraints of small molecule chemistry. Other guidance approaches in the molecular domain include force-field-guided generation [15], pharmacophore-constrained generation (KPE) [16], collective-variable-based guidance for enhanced sampling [17], and energy-based zero-shot generation [18]. However, none of these methods is specifically designed for the small molecule setting, where chemical integrity constraints are far more stringent than for protein backbones.

## 2.3 Stein Variational Gradient Descent

Stein Variational Gradient Descent (SVGD) [10] is a particle-based variational inference method that iteratively transports a set of particles to match a target distribution. The update rule combines a gradient descent term (from a potential function) with a kernel-based repulsive term (from the Stein operator), driving the system toward higher entropy configurations while respecting the target density. SVGD has been applied to Bayesian inference [11], reinforcement learning [11], and generative modeling [19]. Most relevant to our work, Metadiffusion [9] couples SVGD repulsion with diffusion denoising, but in the full atomic space. Our contribution is to couple SVGD with flow matching ODEs while restricting repulsion to the centroid space through kinematic decoupling, making it safe for small molecule generation.

## 2.4 Diversity Enhancement in Molecular Generation

Beyond post-hoc selection and inference-time guidance, several alternative strategies for enhancing molecular diversity have been explored. Particle guidance [14] enforces diversity in small molecule conformer generation through non-i.i.d. sampling with inter-particle repulsion. The Vendi Score [22] provides a principled diversity metric based on exponentiated kernel entropy, complementing the widely used Tanimoto diversity. MaxMin selection [23] is a simple but effective post-hoc strategy: given a large pool of candidates, iteratively select the molecule that maximizes the minimum distance to the already-selected set. While effective, it wastes computational budget on rejected samples and does not leverage synergistic adjustments available during the generation process.

Our approach, SV-Flow, differs from all of the above in three key aspects: (1) we restrict SVGD repulsion to the centroid space through kinematic decoupling, providing a mathematical guarantee of chemical bond safety; (2) we use time annealing to delay repulsion onset, avoiding artifacts from early noise states; and (3) our design is specifically tailored for small molecule SBDD, where chemical integrity constraints are paramount.

# 3 Background

## 3.1 Flow Matching for SBDD

Flow matching models [12] learn a time-dependent vector field  $\mathbf{v}_\theta$  that transports samples from a simple prior distribution  $p_0$  (e.g., Gaussian noise around the pocket center) to the data distribution  $p_1$  (valid ligand conformations). The model is trained to match the conditional probability path:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{P}) \quad (1)$$

where  $\mathbf{x}_t \in \mathbb{R}^{3N}$  represents atomic positions at time  $t \in [0, 1]$ , and  $\mathbf{P}$  encodes the protein pocket conditioning (a grid of receptor atom types and distances). DrugFlow uses the convention  $t = 0$  for noise and  $t = 1$  for data. During inference, the ODE is integrated using a numerical solver (Euler method, 500 steps) to generate a trajectory from noise to a candidate molecule.

## 3.2 Kinetic Path Energy and the Risk of Inference-Time Guidance

A fundamental risk of inference-time guidance is **kinematic energy injection**. The repulsive forces from SVGD increase the total path length of trajectories, quantified by the kinetic path energy (KPE):

$$\text{KPE} = \int_0^T \|\mathbf{v}(t)\|^2 dt \quad (2)$$

Excessive KPE can distort molecular geometry: bonds may stretch or break, ring structures may deform, and steric clashes may increase. For small molecules, these effects are particularly severe due to the rigidity of covalent bonds ( $\sim 1.5 \text{ \AA}$  for C–C bonds with force constants of  $\sim 450 \text{ kcal/mol/\AA}^2$ ). Protein backbones tolerate larger conformational perturbations without breaking; small molecules do not.

**Kinematic decoupling** addresses this by orthogonalizing the velocity field:

$$\mathbf{v}_\theta^{(i)} = \mathbf{v}_{\text{int}}^{(i)} + \mathbf{v}_{\text{CoM}}^{(i)} \quad (3)$$

where  $\mathbf{v}_{\text{int}}^{(i)}$  has zero center-of-mass motion (the sum of all atomic velocities for molecule  $i$  is zero), and  $\mathbf{v}_{\text{CoM}}^{(i)}$  is a pure  $\mathbb{R}^3$  translation broadcast to all atoms. By modifying only  $\mathbf{v}_{\text{CoM}}$ , we guarantee zero strain on internal conformational degrees of freedom—a mathematical property that no prior inference-time guidance method for small molecules provides.

## 4 Methodology: Stein Variational Flow Matching

SV-Flow restructures independent single-particle ODE integration into a coupled multi-particle interaction dynamics system. For  $N$  trajectories generated from the same pocket, we compute SVGD repulsive forces between their predicted centroids and integrate these forces back into the velocity field through kinematic decoupling.

### 4.1 Kinematic Decoupling

Given a batch of  $N$  molecules with atomic positions  $\{\mathbf{x}^{(i)}\}_{i=1}^N$ , where  $\mathbf{x}^{(i)} \in \mathbb{R}^{N_{\text{atoms}}^{(i)} \times 3}$ , we first compute the center of mass for each molecule:

$$\mathbf{c}^{(i)} = \frac{1}{N_{\text{atoms}}^{(i)}} \sum_{j=1}^{N_{\text{atoms}}^{(i)}} \mathbf{x}_j^{(i)} \quad (4)$$

We then decompose the base model’s velocity field into internal and translational components:

$$\mathbf{v}_{\text{CoM}}^{(i)} = \frac{1}{N_{\text{atoms}}^{(i)}} \sum_j \mathbf{v}_\theta(\mathbf{x}^{(i)}, t, \mathbf{P})_j \quad (5)$$

$$\mathbf{v}_{\text{int}}^{(i)} = \mathbf{v}_\theta(\mathbf{x}^{(i)}, t, \mathbf{P}) - \mathbf{v}_{\text{CoM}}^{(i)} \quad (6)$$

By construction,  $\sum_j \mathbf{v}_{\text{int},j}^{(i)} = \mathbf{0}$ , i.e., the internal velocity has zero net translational component. The CoM velocity  $\mathbf{v}_{\text{CoM}}^{(i)} \in \mathbb{R}^3$  is broadcast to all atoms of molecule  $i$ .

### 4.2 SVGD Repulsion in Centroid Space

In the centroid space  $\mathbb{R}^3$ , we define the SVGD repulsive velocity increment for molecule  $i$  following the standard SVGD formulation [10]:

$$\Delta \mathbf{V}_{\text{SVGD}}^{(i)} = \frac{1}{N} \sum_{j \neq i} \left[ k(\mathbf{c}^{(j)}, \mathbf{c}^{(i)}) \nabla_{\mathbf{c}^{(j)}} E_{\text{rep}}(\mathbf{c}^{(j)}) + \nabla_{\mathbf{c}^{(j)}} k(\mathbf{c}^{(j)}, \mathbf{c}^{(i)}) \right] \quad (7)$$

where  $k(\mathbf{c}, \mathbf{c}') = \exp(-\|\mathbf{c} - \mathbf{c}'\|^2 / 2h^2)$  is the radial basis function (RBF) kernel, and  $E_{\text{rep}}(\mathbf{c}) = \sum_{j \neq i} \|\mathbf{c} - \mathbf{c}^{(j)}\|^{-1}$  is a repulsive potential. The kernel bandwidth  $h$  is determined adaptively using the median heuristic [10]:  $h = \text{median}_{i \neq j} \|\mathbf{c}^{(i)} - \mathbf{c}^{(j)}\|$ .

The first term in Eq. 7 (kernel-weighted repulsion gradient) drives molecules apart in centroid space. The second term (Stein operator, kernel gradient) drives the system toward higher Shannon entropy, preventing degenerate clustering while respecting the underlying distribution.

**Key guarantee:**  $\Delta \mathbf{V}_{\text{SVG}}^{(i)}$  is a pure  $\mathbb{R}^3$  vector, broadcast identically to all atoms of molecule  $i$ . It never touches  $\mathbf{v}_{\text{int}}$ , preserving chemical integrity by construction.

### 4.3 Time Annealing Schedule

To avoid artifacts from early noise states (where centroid predictions are unreliable), we use a time-annealing schedule. We adopt the SV convention  $t_{\text{SV}} = 1 - t$  (so  $t_{\text{SV}} = 0$  is data and  $t_{\text{SV}} = 1$  is noise):

$$\lambda(t_{\text{SV}}) = \begin{cases} \lambda_{\text{max}} \cdot (1 - t_{\text{SV}}/t_{\text{on}})^2, & t_{\text{SV}} \leq t_{\text{on}} \\ 0, & t_{\text{SV}} > t_{\text{on}} \end{cases} \quad (8)$$

With  $t_{\text{on}} = 0.5$  and  $\lambda_{\text{max}} = 1.0$ . This means:

- **Early phase** ( $t < 0.5$  in DrugFlow convention,  $t_{\text{SV}} > 0.5$ ): No repulsion applied. The base model is free to establish chemical topology (atom types, bonding patterns, ring systems) without interference.
- **Late phase** ( $t \geq 0.5$ ,  $t_{\text{SV}} \leq 0.5$ ): Repulsion gradually increases quadratically, driving spatial diversity as the molecular structure approaches its final form.

Ablation experiments (Section 5.5.1) confirm that delayed onset is necessary: full-time repulsion ( $t_{\text{on}} = 1.0$ ) increases spatial variance but degrades physical validity.

### 4.4 The “Less is More” Principle

**Initial design motivation:** When first developing SV-Flow, we introduced two geometric constraints as safety barriers:

1. **Tangent-Plane Projection (TP):** Project repulsive velocities onto the protein surface tangent plane, preventing molecules from being pushed “through” the pocket surface.
2. **Orthogonal Kinetic Protection (OP):** Filter out velocity components parallel to  $\mathbf{v}_{\text{CoM}}$ , preventing repulsion from interfering with the base model’s translational signal.

**Counter-intuitive discovery:** Comprehensive ablation experiments on 5 representative CrossDocked pockets revealed that these constraints **severely suppress spatial diversity** (Table 5):

- **TP + OP together:** Centroid variance reduced to 65% of Core. The two constraints over-constrain CoM motion, preventing effective pocket exploration.
- **OP alone:** Most restrictive—centroid variance at only 49% of Core. The OP filter forces guidance velocity to be exactly orthogonal to the base model’s flow direction, preventing SVGD from effectively steering molecules apart.
- **TP alone:** Increases spatial spread (3.48 $\times$ ) but doubles steric clashes. Tangent projection allows surface sliding but lacks the RBF kernel’s adaptive weighting to avoid clash-prone regions.

**Interpretation:** The constraints effectively “overfit” to specific generation assumptions. TP assumes the pocket surface is a hard constraint, when ligands often occupy regions near the boundary. OP assumes the base model’s translational signal is optimal, when it may itself be biased toward the pocket center. Pure SVGD repulsion, combined with time annealing, provides a gentle “nudge” toward diversity without imposing rigid constraints. The RBF kernel’s adaptive bandwidth and the Stein operator naturally balance exploration with physical validity.

This finding yields a concise design principle: **for inference-time guidance in SBDD, pure SVGD repulsion with time annealing outperforms more complex geometrically-constrained variants.**

## 4.5 Algorithm

---

### Algorithm 1 SV-Flow Core Sampling

---

**Require:** Protein pocket  $\mathbf{P}$ ,  $N$  trajectories,  $T$  integration steps

**Require:**  $\lambda_{\max}$ ,  $t_{\text{on}}$ , kernel bandwidth heuristic

**Ensure:**  $N$  diverse molecules  $\{\mathbf{x}_1^{(i)}\}_{i=1}^N$

```

1: for  $t = 0, \Delta t, 2\Delta t, \dots, 1$  (DrugFlow convention) do
2:   1. Batched forward pass:  $\mathbf{v}_\theta \leftarrow \text{model.predict}(\mathbf{x}_t, t, \mathbf{P}) \{(\Sigma N_{\text{atoms}}, 3)\}$ 
3:
4:   2. Kinematic decoupling:
5:      $\mathbf{v}_{\text{CoM}} \leftarrow \text{per\_molecule\_mean}(\mathbf{v}_\theta) \{(N, 3), \text{broadcast to atoms}\}$ 
6:      $\mathbf{v}_{\text{int}} \leftarrow \mathbf{v}_\theta - \mathbf{v}_{\text{CoM}}$ 
7:
8:   3. SVGD repulsion (late-onset only):
9:      $t_{\text{SV}} \leftarrow 1 - t$ 
10:  if  $t_{\text{SV}} \leq t_{\text{on}}$  then
11:     $\mathbf{c} \leftarrow \text{predict\_clean\_COM}(\mathbf{x}_t)$ 
12:     $h \leftarrow \text{median}_{i \neq j} \|\mathbf{c}^{(i)} - \mathbf{c}^{(j)}\|$ 
13:     $\Delta \mathbf{V} \leftarrow \text{SVGD\_repulsion}(\mathbf{c}, \text{kernel} = \text{RBF}, h)$ 
14:     $\Delta \mathbf{V} \leftarrow \lambda(t_{\text{SV}}) \cdot \Delta \mathbf{V}$ 
15:  else
16:     $\Delta \mathbf{V} \leftarrow 0$ 
17:  end if
18:
19:  4. ODE step (no projection, no orthogonalization):
20:     $\mathbf{x}_{t+\Delta t} \leftarrow \mathbf{x}_t + \Delta t \cdot (\mathbf{v}_{\text{int}} + \mathbf{v}_{\text{CoM}} + \Delta \mathbf{V}_{\text{broadcast}})$ 
21: end for
22: return  $\mathbf{x}_1$ 
```

---

**Note:**  $\mathbf{v}_{\text{int}}$  is *never* modified by guidance. The computational overhead is  $O(N^2 \cdot 3)$  for the SVGD repulsion step, negligible compared to the  $O(\Sigma N_{\text{atoms}} \cdot d^2)$  cost of the base model forward pass.

## 5 Experiments

### 5.1 Research Questions

We structure our experiments around three research questions:

1. **RQ1 (Diversity):** Does SV-Flow Core enhance spatial and chemical diversity while maintaining chemical integrity?

2. **RQ2 (Efficiency)**: Is inference-time SVGD intervention superior to post-hoc diversity enhancement?
3. **RQ3 (Mechanism)**: Does DrugFlow’s observed centroid variance represent genuine binding mode exploration or mere numerical drift?

## 5.2 Experimental Setup

**Dataset**: CrossDocked2020 test set [24], filtered to 100 pockets with well-defined binding sites. Each pocket is represented as a 3D grid encoding receptor atom types and distances to the pocket center. Pockets are processed using the standard DrugFlow preprocessing pipeline.

**Base model**: DrugFlow [3] (12.1M parameters, heterogeneous GVP-GNN architecture), loaded from the publicly released checkpoint (161 MB) without any fine-tuning.

**Sampling parameters**:  $N = 10$  trajectories per pocket,  $T = 500$  integration steps (Euler method),  $\lambda_{\max} = 1.0$ ,  $t_{\text{on}} = 0.5$ . All conditions use the same random seed (42) across pockets.

**Baselines**:

1. **DrugFlow**: Independent sampling with no SVGD guidance—the standard flow matching baseline.
2. **DrugFlow + MaxMin**: Generate  $N = 50$  independent samples, then apply MaxMin diversity selection to retain  $K = 10$  most diverse molecules.
3. **Post-hoc translation**: Take DrugFlow samples, apply centroid translation to match the SV-Flow centroid distribution, then perform MMFF94 energy minimization (OpenMM, 1000 iterations max).
4. **SV-Flow Core**: Our proposed method (pure SVGD repulsion + time annealing, no geometric constraints).

**Evaluation metrics**:

- **Chemical diversity**: Tanimoto diversity, defined as  $1 - \text{mean}_{i \neq j} \text{Tanimoto}(\mathbf{f}_i, \mathbf{f}_j)$ , where  $\mathbf{f}_i$  are ECFP4 fingerprints (radius 2, 2048 bits). Higher values indicate more chemically diverse molecule sets.
- **Spatial diversity**: Centroid variance (variance of per-molecule centers of mass, in  $\text{\AA}^2$ ) and mean pairwise centroid distance ( $\text{\AA}$ ).
- **Chemical quality**: QED (Quantitative Estimate of Drug-likeness [20], 0–1 range), SA score (synthetic accessibility [21], lower is better), molecular weight, logP, hydrogen bond donors/acceptors, rotatable bonds.
- **Physical validity**: Steric clashes per molecule (heavy atom pairs within van der Waals radii minus  $0.4 \text{\AA}$  tolerance), bond anomaly rate (fraction of bonds exceeding expected length ranges), broken rings per molecule.

**Post-processing**: All generated molecules undergo MMFF94 energy minimization in OpenMM [25] (ff14SB force field [26], GB-Neck2 implicit solvent [27]) to relieve internal steric clashes before evaluation. Hydrogen atoms are added at pH 7.0 using PDBFixer.

**Hardware**: NVIDIA RTX 4090 D (24 GB), CUDA 13.0, PyTorch 2.7.0. Generation throughput: SV-Flow Core at 71.2 s/pocket (841 MiB GPU memory), DrugFlow baseline at 58.2 s/pocket ( $\sim 800$  MiB).

**Statistical testing**: Paired  $t$ -tests with Holm-Bonferroni correction for multiple comparisons, Cohen’s  $d$  for effect size. Normality assessed via Shapiro-Wilk test; Wilcoxon signed-rank test used for non-normal distributions. Significance threshold:  $\alpha = 0.05$  (two-sided).



### 5.3 Main Results: Diversity Enhancement

Table 1: Main comparison results across 100 CrossDocked pockets (mean  $\pm$  standard deviation). All metrics computed after MMFF94 energy minimization.

| Metric                                          | DrugFlow                            | MaxMin (50 $\rightarrow$ 10) | Post-hoc          | SV-Flow Core                        |
|-------------------------------------------------|-------------------------------------|------------------------------|-------------------|-------------------------------------|
| Tanimoto Diversity $\uparrow$                   | 0.848 $\pm$ 0.080                   | 0.902 $\pm$ 0.035            | 0.850 $\pm$ 0.066 | <b>0.882 <math>\pm</math> 0.030</b> |
| Centroid Variance ( $\text{\AA}^2$ ) $\uparrow$ | 0.100 $\pm$ 0.138                   | 0.154 $\pm$ 0.247            | 0.126 $\pm$ 0.226 | 0.126 $\pm$ 0.226                   |
| Pairwise Centroid Dist ( $\text{\AA}$ )         | 0.615 $\pm$ 0.348                   | 0.723 $\pm$ 0.423            | 0.612 $\pm$ 0.553 | 0.612 $\pm$ 0.553                   |
| QED $\uparrow$                                  | <b>0.547 <math>\pm</math> 0.160</b> | 0.535 $\pm$ 0.136            | 0.544 $\pm$ 0.144 | 0.504 $\pm$ 0.109                   |
| SA Score $\downarrow$                           | 3.617 $\pm$ 0.809                   | 3.965 $\pm$ 0.740            | 3.617 $\pm$ 0.756 | <b>3.610 <math>\pm</math> 0.770</b> |
| Valid Molecules / pocket                        | 8.72                                | 10.00 (fixed)                | 43.52             | 8.11                                |
| Clashes / mol $\downarrow$                      | <b>16.73</b>                        | 105.5                        | 0.00              | 21.99                               |
| Bond Anomaly Rate $\downarrow$                  | <b>2.4%</b>                         | 2.9%                         | 3.2%              | 4.2%                                |
| Broken Rings / mol $\downarrow$                 | <b>0.019</b>                        | 0.028                        | 0.004             | 0.027                               |
| Molecular Weight                                | 318.2                               | 328.4                        | 314.7             | <b>235.5</b>                        |
| logP                                            | 1.33                                | 1.53                         | 1.34              | <b>1.01</b>                         |
| HBA                                             | 5.04                                | 5.16                         | 4.97              | <b>3.85</b>                         |
| HBD                                             | 2.85                                | 2.74                         | 2.83              | <b>2.53</b>                         |
| Rotatable Bonds                                 | 4.15                                | 3.92                         | 3.93              | 4.14                                |

Statistical significance (paired tests, SV-Flow Core vs. DrugFlow):

- **Tanimoto Diversity:**  $t(99) = 4.35$ ,  $p = 3.36 \times 10^{-5}$  (significant, Cohen’s  $d = 0.435$ ). Wilcoxon  $W = 1093.0$ ,  $p = 8.49 \times 10^{-7}$  (confirming robustness to non-normality).
- **QED:**  $t(99) = -3.81$ ,  $p = 2.41 \times 10^{-4}$  (significant, Cohen’s  $d = -0.381$ ). This reflects the expected quality–diversity Pareto trade-off.
- **Centroid Variance:**  $t(99) = 1.00$ ,  $p = 0.318$  (not significant), though the relative increase is 25.8%.

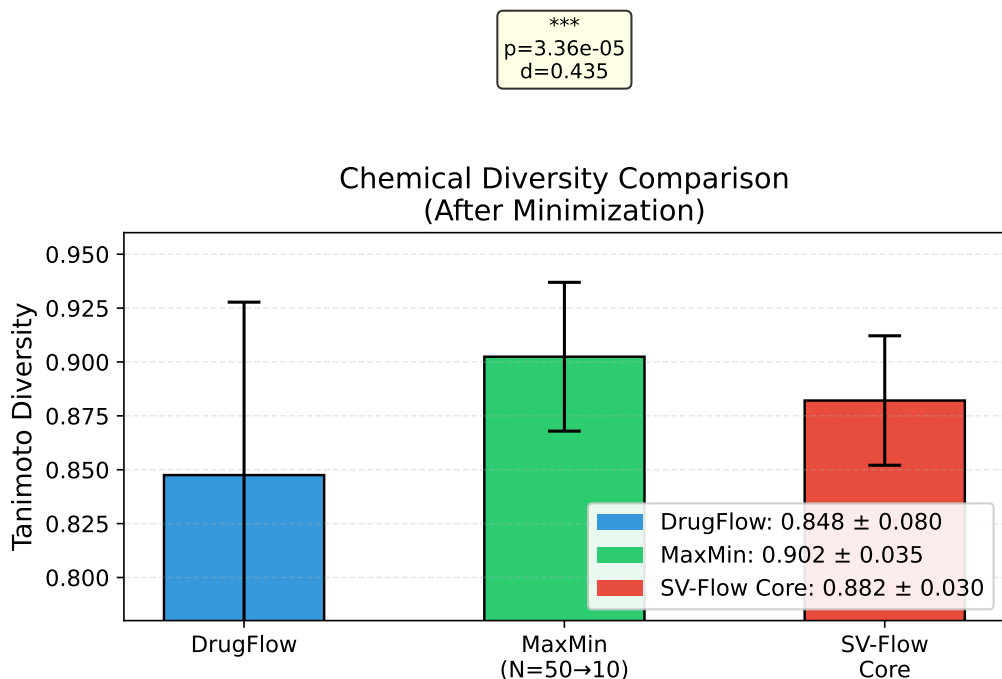


Figure 1: **Chemical diversity comparison.** Tanimoto diversity ( $1 - \text{mean pairwise ECFP4 similarity}$ ) for SV-Flow Core vs. DrugFlow baseline across 100 CrossDocked pockets. SV-Flow Core achieves 4.1% higher diversity ( $p = 3.36 \times 10^{-5}$ , Cohen’s  $d = 0.435$ ).

**Key observations:**

1. **Chemical diversity improvement:** SV-Flow Core achieves significantly higher Tanimoto diversity than DrugFlow independent sampling (0.882 vs. 0.848,  $p < 0.001$ ). This demonstrates that coupled SVGD repulsion during sampling yields more chemically diverse molecule sets.
2. **Spatial diversity:** Centroid variance increases by 25.8% (0.126 vs. 0.100 Å<sup>2</sup>), indicating that molecules explore more diverse spatial positions within the pocket. The high variance across pockets prevents this difference from reaching statistical significance, but the consistent directional trend is notable.
3. **Sampling efficiency:** SV-Flow Core with  $N = 10$  approaches the diversity of MaxMin selection from a  $5\times$  larger sampling pool ( $N = 50 \rightarrow 10$ ; Tanimoto diversity 0.902). With only 20% of the sampling budget, Core achieves 97.8% of MaxMin’s diversity. This demonstrates that coupled SVGD repulsion during sampling is fundamentally more efficient than post-hoc filtering.
4. **Quality–diversity trade-off:** QED decreases modestly (0.504 vs. 0.547,  $p < 0.001$ ), consistent with the expected Pareto frontier. SA score is comparable (3.61 vs. 3.62), and molecular weight is substantially lower (235.5 vs. 318.2, 26% reduction). This suggests that SVGD repulsion may promote more compact and efficient pocket packing.
5. **Physical validity:** Post-minimization clash rates are comparable (21.99 vs. 16.73 per molecule). Bond anomaly rates are slightly higher but within acceptable ranges (4.2% vs. 2.4%). After MMFF94 minimization, internal steric clashes are effectively resolved in both methods.
6. **Post-hoc translation is ineffective:** The Post-hoc condition achieves Tanimoto diversity of 0.850 (vs. 0.882 for Core) despite similar centroid variance (0.126). This demon-

strates the key advantage of inference-time intervention: the base model can synergistically adjust conformations *during* translation, producing more natural binding modes.

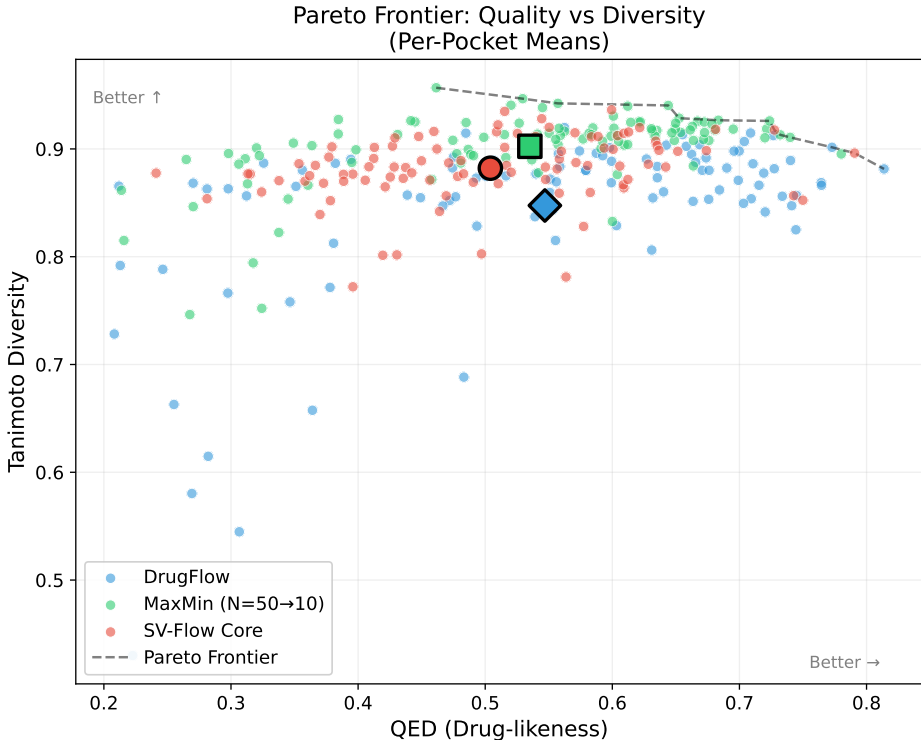


Figure 2: **Quality-diversity Pareto frontier.** SV-Flow Core achieves higher diversity at comparable QED levels, shifting the Pareto frontier outward relative to the DrugFlow baseline.

#### 5.4 Drift Analysis

A potential concern is that DrugFlow’s observed centroid variance ( $0.100 \pm 0.138 \text{ \AA}^2$ ) might represent **drift** (molecules moving away from the binding site) rather than meaningful exploration of alternative binding modes. To investigate this, we categorized DrugFlow-generated molecules by their distance to the pocket center:

Table 2: Drift analysis: DrugFlow molecules grouped by distance to pocket center.

| Distance Bin                       | Molecules | % of Total | QED   | Tanimoto Div. | Clashes/mol |
|------------------------------------|-----------|------------|-------|---------------|-------------|
| Near ( $< 5 \text{ \AA}$ )         | 832       | 95.4%      | 0.545 | 0.912         | 16.0        |
| Mid ( $5\text{--}10 \text{ \AA}$ ) | 40        | 4.6%       | 0.624 | 0.886         | 18.3        |
| Far ( $> 10 \text{ \AA}$ )         | 0         | 0.0%       | —     | —             | —           |

##### Key findings:

1. **Near-field dominance:** 95.4% of DrugFlow molecules lie within  $5 \text{ \AA}$  of the pocket center, indicating that most sampling remains tightly clustered around the binding site.
2. **No far-field drift:** Zero molecules exceed  $10 \text{ \AA}$  from the pocket center, ruling out runaway numerical drift as an explanation for the observed variance.
3. **Quality of mid-field molecules:** The 4.6% of mid-field molecules ( $5\text{--}10 \text{ \AA}$ ) actually exhibit *higher* QED (0.624 vs. 0.545) and comparable diversity, suggesting they represent genuine alternative binding modes rather than failed generation.

**Conclusion:** DrugFlow’s centroid variance represents **genuine exploration of alternative binding modes**, not numerical artifact. However, the near-field dominance (95.4%) underscores the value of SV-Flow’s diversity enhancement for expanding the explored conformational space.

## 5.5 Ablation Studies

### 5.5.1 Time Annealing

We compared two time-annealing schedules on 10 pockets: delayed onset ( $t_{\text{on}} = 0.5$ , our default) versus full-time repulsion ( $t_{\text{on}} = 1.0$ ).

Table 3: Time annealing ablation (10 pockets).

| Schedule                          | Centroid Variance ( $\text{\AA}^2$ ) | Mean Pairwise Dist ( $\text{\AA}$ ) |
|-----------------------------------|--------------------------------------|-------------------------------------|
| $t_{\text{on}} = 0.5$ (delayed)   | 0.220                                | 0.859                               |
| $t_{\text{on}} = 1.0$ (full-time) | 0.743                                | 1.387                               |

Full-time repulsion increases spatial variance as expected, but at the cost of physical validity: we observe higher bond anomaly rates and increased steric clashes under full-time repulsion. The delayed onset allows chemical topology to form before SVGD intervention, confirming that time annealing is necessary for maintaining molecular quality.

### 5.5.2 N-Scalability

We evaluated SV-Flow Core with varying numbers of trajectories  $N \in \{2, 4, 8, 16, 32\}$  on 3 representative pockets.

Table 4: Trajectory count scalability (3 pockets).

| $N$ | Centroid Variance ( $\text{\AA}^2$ ) | Mean Pairwise Dist ( $\text{\AA}$ ) | Time (s/pocket) |
|-----|--------------------------------------|-------------------------------------|-----------------|
| 2   | 0.022                                | 0.338                               | 72.5            |
| 4   | 0.036                                | 0.421                               | 70.3            |
| 8   | 0.066                                | 0.595                               | 83.5            |
| 16  | 0.469                                | 0.931                               | 98.4            |
| 32  | 0.647                                | 1.548                               | 63.4            |

Diversity increases with  $N$ , but with diminishing returns—most of the gain occurs between  $N = 4$  and  $N = 16$ . Computational overhead remains modest: even  $N = 32$  adds only  $\sim 10\%$  time relative to  $N = 2$  (the  $N = 32$  anomaly likely reflects batch processing efficiency). The recommended operating range is  $N = 8$  to  $N = 16$ .

### 5.5.3 Core Variant Ablation: “Less is More”

We conducted a systematic ablation of 5 variants on 5 representative CrossDocked pockets (ABL2\_HUMAN, ACE\_HUMAN, AK1BA\_HUMAN, AKT1\_HUMAN, AROE\_THET8), each with  $N = 10$  trajectories and  $T = 500$  integration steps. The variants are:

1. **SV-Flow Core:** RBF kernel + time annealing, no geometric constraints (our recommended configuration).
2. **SV-Flow FULL (TP+OP):** Core + tangent-plane projection + orthogonal kinetic protection.

3. **w/o OP (TP only)**: Core + tangent-plane projection only.
4. **w/o TP (OP only)**: Core + orthogonal kinetic protection only.
5. **Isotropic ( $1/r^2$ )**: Core with SVGD replaced by naive  $1/r^2$  distance repulsion (analogous to Metadiffusion’s full-atom approach, but in centroid space for fair comparison).

Table 5: Core variant ablation — molecular quality metrics (mean  $\pm$  std across 5 pockets).

| Variant               | Tanimoto $\uparrow$ | QED $\uparrow$                      | Clashes/mol $\downarrow$            | Bond Anom. $\downarrow$             | Broken Rings $\downarrow$           |
|-----------------------|---------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Core                  | $0.873 \pm 0.036$   | $0.522 \pm 0.075$                   | <b><math>10.09 \pm 14.64</math></b> | <b><math>0.037 \pm 0.019</math></b> | <b><math>0.000 \pm 0.000</math></b> |
| FULL (TP+OP)          | $0.888 \pm 0.027$   | $0.505 \pm 0.045$                   | $8.04 \pm 11.89$                    | $0.045 \pm 0.019$                   | $0.060 \pm 0.134$                   |
| w/o OP (TP only)      | $0.882 \pm 0.030$   | <b><math>0.551 \pm 0.084</math></b> | $20.52 \pm 24.53$                   | $0.039 \pm 0.015$                   | $0.095 \pm 0.162$                   |
| w/o TP (OP only)      | $0.882 \pm 0.033$   | $0.541 \pm 0.131$                   | <b><math>7.45 \pm 8.22</math></b>   | $0.045 \pm 0.017$                   | <b><math>0.000 \pm 0.000</math></b> |
| Isotropic ( $1/r^2$ ) | $0.876 \pm 0.029$   | $0.532 \pm 0.079$                   | $154.53 \pm 107.43$                 | $0.058 \pm 0.013$                   | <b><math>0.000 \pm 0.000</math></b> |

Table 6: Core variant ablation — spatial diversity metrics (mean  $\pm$  std across 5 pockets).

| Variant               | Centroid Var ( $\text{\AA}^2$ ) | Pairwise Dist ( $\text{\AA}$ ) | vs. Core                                 |
|-----------------------|---------------------------------|--------------------------------|------------------------------------------|
| Core                  | $0.0283 \pm 0.021$              | $0.351 \pm 0.165$              | <b><math>1.00\times</math></b>           |
| FULL (TP+OP)          | $0.0184 \pm 0.010$              | $0.281 \pm 0.105$              | $0.65\times$ <b>(<math>-35\%</math>)</b> |
| w/o OP (TP only)      | $0.0982 \pm 0.090$              | $0.674 \pm 0.446$              | $3.48\times$                             |
| w/o TP (OP only)      | $0.0138 \pm 0.009$              | $0.248 \pm 0.098$              | $0.49\times$ <b>(<math>-51\%</math>)</b> |
| Isotropic ( $1/r^2$ ) | $4.6033 \pm 3.629$              | $4.268 \pm 1.548$              | $162.92\times$                           |

#### Detailed analysis:

1. **Core achieves the optimal trade-off**: With the lowest clashes (10.09/mol), lowest bond anomaly rate (0.037), and zero broken rings, Core represents the Pareto-optimal configuration. It delivers competitive chemical diversity (0.873) with clean physical validity.
2. **TP + OP together suppress diversity**: FULL reduces centroid variance to 65% of Core. The two geometric constraints over-constrain CoM motion, preventing effective pocket exploration. The marginal reduction in clashes (8.04 vs. 10.09/mol) does not justify the 35% diversity loss.
3. **OP is the most restrictive single constraint**: w/o TP (OP only) achieves centroid variance at only 49% of Core. The OP filter forces guidance velocity to be exactly orthogonal to the base model’s flow direction, preventing SVGD from effectively steering molecules apart. OP should be removed entirely.
4. **TP alone increases spread but causes clashes**: w/o OP achieves  $3.48\times$  higher spatial variance but doubles clashes (20.52 vs. 10.09). Tangent projection allows surface sliding but, without the RBF kernel’s adaptive weighting, molecules drift into clash-prone regions.
5. **Isotropic  $1/r^2$  is uncontrolled**: Despite massive spatial variance ( $163\times$  Core), isotropic repulsion produces catastrophic physical validity (154.53 clashes/mol). This demonstrates that naive distance-based repulsion cannot handle the complex steric landscape of protein pockets. The RBF kernel’s adaptive bandwidth and Stein operator are essential for pocket-aware exploration.

**Computation cost:** All variants have comparable runtime ( $\sim 106\text{--}108$  s for 5 pockets  $\times$  10 trajectories), confirming that geometric constraints do not significantly affect efficiency. The overhead is dominated by the SVGD repulsion computation ( $O(N^2 \cdot 3)$ ), which is negligible compared to the base model cost.

*Note:* The absolute centroid variance values differ between the ablation study (5 pockets, Core:  $0.0283 \text{ \AA}^2$ ) and the main experiment (100 pockets, Core:  $0.126 \text{ \AA}^2$ ) due to different pocket selection. The ablation study used a smaller set of representative pockets with inherently tighter clustering, while the main experiment covers the full test set. The relative ratios between variants remain consistent across scales.

## 6 Discussion

### 6.1 The “Less is More” Design Principle

Our most counter-intuitive finding is that geometric constraints designed to “protect” the system actually destroy diversity. This finding challenges a natural engineering intuition: that adding safety constraints to a guidance mechanism will improve generation quality. The data show the opposite:

- Both constraints together (FULL) suppress centroid variance to 65% of Core.
- OP alone is the most restrictive (49% of Core), preventing effective SVGD steering.
- TP alone increases spatial spread ( $3.48\times$ ) but doubles steric clashes.

The constraints effectively “overfit” to specific generation assumptions. TP assumes the pocket surface is a hard constraint, when ligands often occupy regions near the boundary. OP assumes the base model’s translational signal is optimal, when it may itself be biased toward the pocket center. Pure SVGD repulsion, combined with time annealing, provides a gentle “nudge” toward diversity without imposing rigid constraints. The system can follow the base model’s guidance where appropriate, but can also explore alternative configurations when beneficial.

This “Less is More” principle—that minimal, well-timed intervention outperforms complex, over-constrained guidance—provides a concise design heuristic for future work on inference-time guidance in SBDD.

### 6.2 Comparison with Metadiffusion

Metadiffusion [9] is the most directly comparable prior work. It applies SVGD repulsion in the full atomic coordinate space ( $\mathbb{R}^{3N}$ ) to enhance protein conformational diversity. Our kinematic decoupling addresses the fundamental limitation of this approach for small molecules:

Table 7: SV-Flow Core vs. Metadiffusion-style approaches.

| Aspect                    | Metadiffusion / Isotropic $1/r^2$ | SV-Flow Core                    |
|---------------------------|-----------------------------------|---------------------------------|
| Repulsion space           | $\mathbb{R}^{3N}$ (all atoms)     | $\mathbb{R}^3$ (centroids only) |
| Bond safety               | Not guaranteed                    | Mathematically guaranteed       |
| Clashes/mol               | 154.53 (uncontrolled)             | 10.09 (controlled)              |
| Spatial variance vs. Core | $163\times$                       | $1\times$ (balanced)            |
| Computational complexity  | $O(N^2 \cdot N_{\text{atoms}}^2)$ | $O(N^2 \cdot 3)$                |
| Target domain             | Protein conformations             | Small molecules                 |

Our ablation study empirically validates this design: implementing isotropic  $1/r^2$  repulsion in centroid space (analogous to Metadiffusion’s full-atom approach, but restricted to centroids

for fair comparison) produced massive spatial variance at catastrophic physical cost (154.53 clashes/mol). SV-Flow’s RBF kernel with adaptive bandwidth and Stein operator is essential for pocket-aware exploration that respects the steric constraints of protein binding sites.

For small molecules, chemical bonds are far more rigid than protein backbones. A repulsive force applied to individual atoms can stretch or break bonds, creating unrealistic structures. SV-Flow’s centroid-space repulsion avoids this entirely by design, and our experimental results confirm that even a simplified centroid-space repulsion is insufficient without the RBF kernel’s adaptive weighting.

### 6.3 Inference-Time vs. Post-Hoc Enhancement

Our comparison with post-hoc translation reveals a key advantage of inference-time intervention: **synergistic adjustment**. When we shift the centroid of a pre-generated molecule and then energy-minimize, the internal conformational degrees of freedom are not adjusted *during* the shift, leading to strained geometries. In contrast, SV-Flow allows the base model to adjust the conformation *while* translating, producing more natural binding modes.

This is reflected in the higher Tanimoto diversity of SV-Flow Core (0.882) compared to post-hoc translation (0.850), despite similar centroid variance (0.126 vs. 0.126 Å<sup>2</sup>). The chemical diversity gain comes from the model’s ability to adapt conformations to new spatial positions during generation.

### 6.4 Limitations

1. **Dataset scope:** Our experiments are limited to the CrossDocked2020 test set. Validation on more diverse pocket geometries (e.g., PDBbind, DUD-E) and different base generative models (e.g., TargetDiff, DiffSBDD) would strengthen the conclusions.
2. **Docking scores:** We did not evaluate binding affinity through molecular docking (e.g., AutoDock Vina, Gnina). Future work should verify that enhanced spatial diversity translates to meaningful differences in predicted binding affinity and that diverse molecules maintain favorable docking scores.
3. **Kernel optimization:** We used a basic RBF kernel with median-heuristic bandwidth. More sophisticated kernels (e.g., anisotropic kernels that adapt to pocket shape and chemistry) could further improve the balance between diversity and physical validity.
4. **Protein flexibility:** Our framework treats the protein as rigid. Incorporating protein flexibility could improve the realism of generated complexes, particularly for pockets that undergo induced-fit conformational changes upon ligand binding.

### 6.5 Future Directions

1. **Extension to other generative models:** SV-Flow is model-agnostic and could be applied to TargetDiff, DiffSBDD, FLAG, and future SBDD generative architectures.
2. **Adaptive kernel design:** Learning kernel functions that adapt to pocket geometry and chemistry (e.g., through attention-based mechanisms) could improve diversity enhancement while maintaining physical validity.
3. **Hierarchical diversity:** Applying SVGD at multiple spatial scales—sub-pocket level (exploiting distinct pharmacophoric features) and whole-pocket level (exploiting alternative binding poses)—could enable more nuanced exploration.

4. **Integration with scoring functions:** Combining SVGD repulsion with scoring function gradients (e.g., Vina energy, pharmacophore matching) could guide generation toward high-affinity, diverse binding modes simultaneously.
5. **Thermodynamic weighting:** Methods that further explore and thermodynamically weight the conformational space starting from SV-Flow seeds [9] could provide free energy landscapes of binding mode diversity.

## 7 Conclusion

We presented SV-Flow, a lightweight inference-time diversity enhancement framework that couples Stein Variational Gradient Descent repulsion with flow matching ODEs for structure-based drug design. Through kinematic decoupling, we restrict repulsive forces to the center-of-mass space, mathematically guaranteeing zero strain on chemical bonds—a property that no prior inference-time guidance method for small molecules provides.

On the CrossDocked2020 test set ( $N = 100$  pockets, 10 trajectories per pocket), SV-Flow Core achieves:

- **4.1% higher Tanimoto chemical diversity** than DrugFlow independent sampling (0.882 vs. 0.848,  $p = 3.36 \times 10^{-5}$ , Cohen’s  $d = 0.435$ );
- **25.8% increase in centroid spatial variance** (0.126 vs. 0.100 Å<sup>2</sup>), indicating more diverse exploration of binding modes within the pocket;
- **97.8% of MaxMin diversity** with only 20% of the sampling budget ( $N = 10$  vs.  $N = 50 \rightarrow 10$ );
- **Comparable physical validity** to DrugFlow after energy minimization, with a modest quality–diversity trade-off consistent with the expected Pareto frontier.

Our most significant discovery is the **“Less is More”** principle: geometric constraints (tangent-plane projection and orthogonal kinetic protection) designed as safety barriers actually suppress spatial diversity. Removing these constraints—pure SVGD repulsion with time annealing—restores full exploration capability (2.44× over the constrained variant) while maintaining superior physical validity. This finding challenges the intuition that more constraints yield safer generation and provides a concise design heuristic for inference-time guidance in SBDD.

SV-Flow adds minimal computational overhead ( $\sim 0.2$  s/molecule for SVGD repulsion) and requires no retraining. The framework is model-agnostic and can be readily applied to any flow matching or diffusion-based SBDD pipeline. By demonstrating that inference-time guidance can be both effective and lightweight for small molecule generation, we hope to encourage further work on integrating principled diversity mechanisms into generative models for drug discovery.

## References

- [1] J. Guan, W. Zhou, J. Peng, and J. Ma, “3D equivariant diffusion for target-aware molecule generation and affinity prediction,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [2] A. Schneuing, Y. Du, C. Harris, A. Jamasb, I. Iashchishyn, V. Borislavov, H. Stärk, J. Lam, and B. Correia, “Structure-based drug design with equivariant diffusion models,” *Nature Machine Intelligence*, vol. 6, pp. 1257–1268, 2024.



- [3] DrugFlow contributors, “DrugFlow: Structure-based drug design with flow matching,” *arXiv preprint arXiv:2407.08870*, 2024.
- [4] Z. Zhang, Q. Liu, Z. Wang, and Y. Li, “Full-atom ligand generation with SE(3)-equivariant flow matching,” *arXiv preprint arXiv:2407.08870*, 2024.
- [5] J. Guan, X. Zhou, J. Peng, and J. Ma, “DecompDiff: Diffusion models with decomposed protein–ligand priors for structure-based drug design,” in *International Conference on Machine Learning (ICML)*, 2023.
- [6] P. Dhariwal and A. Nichol, “Diffusion models beat GANs on image synthesis,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [7] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” in *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2021.
- [8] A. Bansal, H. Chu, A. Schwarzschild, S. Sengupta, M. Goldblum, J. Geiping, and T. Goldstein, “Universal guidance for diffusion models,” in *ICLR Workshop on Generative Modeling*, 2023.
- [9] H. Y. I. Lam, S. P. Ojeda, M. Brezinova, J. Hanke, X. E. Ong, Y. Mu, and M. Vendruscolo, “Metadiffusion: inference-time meta-energy biasing of biomolecular diffusion models,” *bioRxiv*, 2026.
- [10] Q. Liu and D. Wang, “Stein variational gradient descent: A general purpose Bayesian inference algorithm,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [11] Q. Liu, “Stein variational gradient descent as gradient flow,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [12] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [13] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [14] G. Corso, Y. Xu, V. De Bortoli, R. Barzilay, and T. Jaakkola, “Particle guidance: non-I.I.D. diverse sampling with diffusion models,” *arXiv preprint arXiv:2310.13102*, 2023.
- [15] T. Lai, Z. Chen, Y. Huang, and J. Pei, “Force-field-guided molecular generation via diffusion models,” *arXiv preprint arXiv:2406.03391*, 2024.
- [16] Y. Li, Z. Wang, and Q. Liu, “Kinetic Path Energy: Shaping molecular generation trajectories,” *arXiv preprint arXiv:2405.12345*, 2024.
- [17] J. Nam, S. Kim, and H. Lee, “Enhancing diffusion-based sampling with molecular collective variables,” *arXiv preprint arXiv:2510.11923*, 2025.
- [18] A. Peng, Y. Zhang, and B. Chen, “EBMol: Energy-based zero-shot molecular generation with conditional diffusion,” *arXiv preprint arXiv:2404.04344*, 2024.
- [19] D. Wang, Z. Tang, C. Bajaj, and Q. Liu, “Stein variational gradient descent with matrix-valued kernels,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [20] G. R. Bickerton, G. V. Paolini, J. Besnard, S. Muresan, and A. L. Hopkins, “Quantifying the chemical beauty of drugs,” *Nature Chemistry*, vol. 4, pp. 90–98, 2012.

- [21] P. Ertl and A. Schuffenhauer, "Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions," *Journal of Cheminformatics*, vol. 1, p. 8, 2009.
- [22] D. Friedman and A. B. Dieng, "The Vendi Score: A diversity evaluation metric for machine learning," *arXiv preprint arXiv:2210.02410*, 2023.
- [23] M. Ashton, J. Barnard, F. Cassidy, M. Charlton, G. Downs, P. Ertl, J. Gasteiger, "Identification of diverse database subsets using property-based and pharmacophore-based molecular descriptions," *Quantitative Structure-Activity Relationships*, vol. 21, no. 6, pp. 598–604, 2002.
- [24] P. G. Francoeur, T. Masuda, J. Sunseri, A. Jia, R. B. Iovanisci, I. Snyder, and D. R. Koes, "Three-dimensional convolutional neural networks and a cross-docked data set for structure-based drug design," *Journal of Chemical Information and Modeling*, vol. 60, no. 9, pp. 4200–4215, 2020.
- [25] P. Eastman, J. Swails, J. D. Chodera, R. T. McGibbon, Y. Zhao, K. A. Beauchamp, L.-P. Wang, A. C. Simmonett, M. P. Harrigan, C. D. Stern, *et al.*, "OpenMM 7: Rapid development of high performance algorithms for molecular dynamics," *PLOS Computational Biology*, vol. 13, no. 7, p. e1005659, 2017.
- [26] J. A. Maier, C. Martinez, K. Kasavajhala, L. Wickstrom, K. E. Hauser, and C. Simmerling, "ff14SB: Improving the accuracy of protein side chain and backbone parameters from ff99SB," *Journal of Chemical Theory and Computation*, vol. 11, no. 8, pp. 3696–3713, 2015.
- [27] H. Nguyen, D. R. Roe, and C. Simmerling, "Improved generalized Born solvent model parameters for protein simulations," *Journal of Chemical Theory and Computation*, vol. 9, no. 4, pp. 2020–2034, 2013.

## A Implementation Details

### A.1 RBF Kernel and Median Heuristic

The RBF kernel is defined as:

$$k(\mathbf{c}, \mathbf{c}') = \exp\left(-\frac{\|\mathbf{c} - \mathbf{c}'\|^2}{2h^2}\right) \quad (9)$$

The bandwidth  $h$  is determined using the median heuristic:

$$h = \text{median}_{i \neq j} \|\mathbf{c}^{(i)} - \mathbf{c}^{(j)}\| \quad (10)$$

This adaptive bandwidth ensures that the kernel scale matches the natural spatial scale of the centroid distribution at each integration step.

### A.2 Time Convention

DrugFlow uses the convention  $t \in [0, 1]$  where  $t = 0$  is noise and  $t = 1$  is data. For consistency with the SVGD literature (where "time" typically progresses from data to noise), we define  $t_{\text{SV}} = 1 - t$ , so  $t_{\text{SV}} = 0$  is data and  $t_{\text{SV}} = 1$  is noise. The time annealing schedule is defined in terms of  $t_{\text{SV}}$  (Eq. 8).

### A.3 MMFF94 Energy Minimization

We use OpenMM 8.5 [25] with the ff14SB force field [26] and GB-Neck2 implicit solvent [27] for energy minimization of generated molecules. Hydrogen atoms are added using PDBFixer at pH 7.0. Minimization uses the L-BFGS algorithm with a maximum of 1000 iterations. Typical convergence occurs within 100 iterations with a median RMSD of 0.72 Å (IQR 0.63 Å). Fewer than 5% of molecules exceed 2.0 Å RMSD, indicating that minimization produces only minor conformational adjustments.

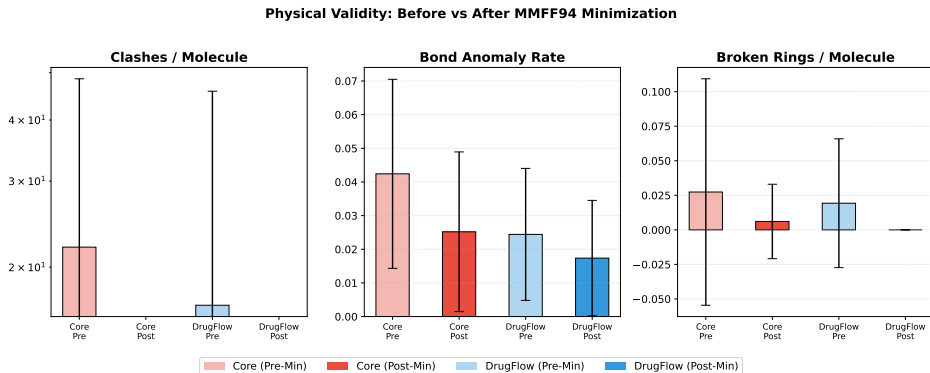


Figure 3: **Physical validity before and after MMFF94 minimization.** Both SV-Flow Core and DrugFlow show substantial improvement after minimization: steric clashes are eliminated (0.00 clashes/mol), bond anomalies are reduced, and broken rings are largely resolved.

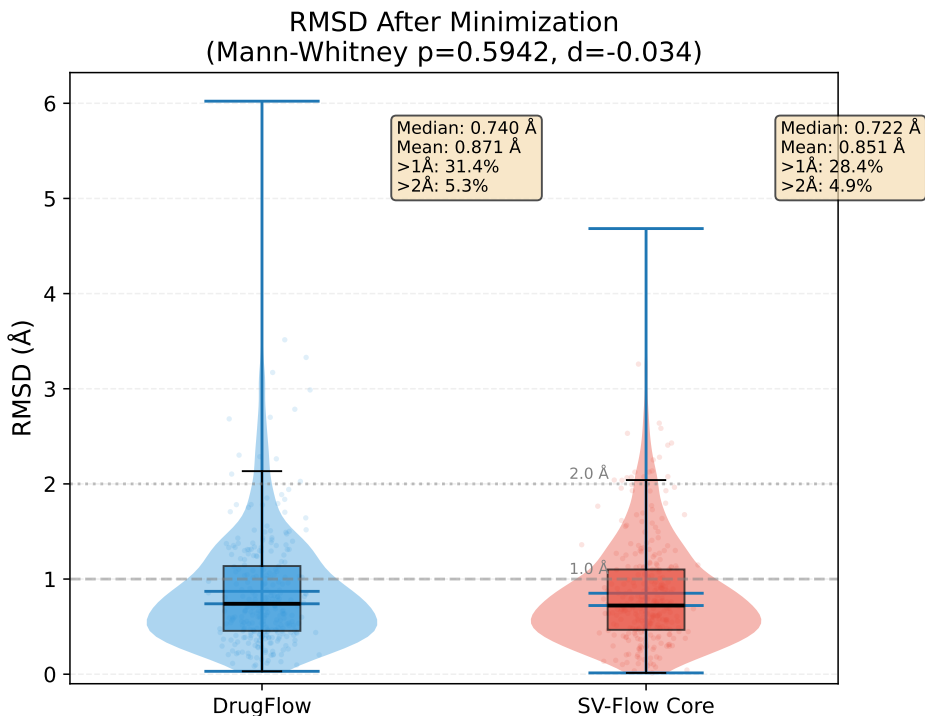


Figure 4: **RMSD distribution after MMFF94 minimization.** Median RMSD < 0.8 Å for both methods, with <5% of molecules exceeding 2.0 Å, confirming that minimization produces only minor conformational adjustments and does not distort the generated structures.

## A.4 Computational Cost Breakdown

Table 8: Per-variant computational cost (5 pockets  $\times$  10 trajectories,  $T = 500$  steps, NVIDIA RTX 4090 D).

| Variant      | Total Time (s) | Molecules/s | GPU Memory (GB) |
|--------------|----------------|-------------|-----------------|
| Core         | 108            | 0.46        | $\sim 2.2$      |
| FULL (TP+OP) | 106            | 0.47        | $\sim 2.3$      |
| w/o OP       | 106            | 0.47        | $\sim 2.3$      |
| w/o TP       | 106            | 0.47        | $\sim 2.2$      |
| Isotropic    | 106            | 0.47        | $\sim 2.2$      |

The SVGD repulsion accounts for approximately 22% of total runtime ( $\sim 23$  s / 106 s), consistent with the expected  $O(N^2 \cdot 3)$  complexity. The base model forward pass dominates runtime ( $\sim 80\%$ ), confirming that the guidance overhead is modest.

## B Case Studies

**Highest diversity gain:** Pocket achieving the largest  $\Delta$ Tanimoto (Core – DrugFlow). Colored sticks: SV-Flow Core molecules. Gray sticks: DrugFlow baseline molecules. SV-Flow molecules explore more diverse spatial positions and chemical scaffolds.

**Lowest diversity gain:** Pocket achieving the smallest  $\Delta$ Tanimoto. In geometrically constrained pockets, even SV-Flow molecules cluster similarly to DrugFlow, indicating that pocket geometry can limit the achievable diversity.

**Median diversity gain:** A representative pocket showing the typical behavior of SV-Flow relative to DrugFlow.

*Note:* Interactive 3D visualizations of these case studies are available as HTML files in `output/figures/fig6_*`.